CAST: A Cascade Spectral-Aware Transformer for Hyperspectral Image Change Detection

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***Abstract*— Hyperspectral image change detection (HSI-CD) aims to detect subtle changes on the Earth's surface through approximately continuous spectral information, which has gradually become a very important research hotspot in the field of remote sensing (RS). In recent years, convolutional neural networks (CNNs)-based HSI-CD methods have shown strong feature extraction capabilities. However, due to the simple fusion of spectral information in the channel dimension by CNN, the medium- and long-term sequence properties of spectral features cannot be well mined and represented. Most previous studies mainly extract semantic features from images at different times, ignoring the temporal correlation between features, which cannot fully extract and effectively utilize temporal–spatial–spectral features. To this end, this article proposes a cascade spectral-aware transformer (CAST) for HSI-CD. First, we propose a temporal–spatial transformer (TS-Former) to enhance the temporal correlation and spatial global relationship of extracted features, thereby addressing the insufficient consideration of temporal correlation. Second, a spectral-awareness transformer (SA-Former) is designed to better mine and represent the sequence properties of spectral features, especially the medium- and long-term dependencies. Finally, we observe a spectral distortion in the process of extracting temporal–spatial features and, based on this, present a spectral constraint module (SCM) to preserve the sequence properties of spectral features and reduce the distortion of the spectrum. Extensive experiments on three challenging hyperspectral datasets demonstrate that our method achieves state-of-the-art results. The code is available at https://github.com/tianshunli/CAST.**

***Index Terms*— Change detection (CD), hyperspectral image (HSI), spectral constraint module (SCM), transformer.**

I. Introduction

With the rapid development of remote sensing (RS) technology, RS images have become a commonly used source for monitoring the Earth's surface [1]. Change detection (CD) is the process of detecting change information and change areas of two or more images acquired at different times at the same location [2] and is widely used in various fields, such as land use and land cover change [3], forest and vegetation change monitoring [4], ecosystem monitoring, urban sprawl studies [5], resource management, and damage assessment [6]. Due to the near-continuous spectral properties of hyperspectral images (HSIs) [7], HSI-CD has the advantages of high detection accuracy, and few false and missed detection phenomena. Therefore, HSI-CD is gradually becoming a very important research topic in the field of RS.

In general, traditional HSI-CD methods can be implemented in three steps: 1) image preprocessing, which includes geometric correction, radiometric correction, image registration, and denoising operations on the initial image [8], to reduce errors in difference image generation; 2) generation of difference images, which generates changed regions in HSI; and 3) evaluation, which measures the results of HSI-CD and evaluates the performance of algorithms.

Change vector analysis (CVA) is a classical algebra-based method [9], that generates a difference map by calculating the Euclidean distance between two image pixels and then clusters the difference maps to obtain the final change results. Although these methods are relatively simple, straightforward, and easy to implement and interpret, they also have drawbacks, such as low accuracy in multiclass CD and difficulty in choosing appropriate thresholds [10]. Transformation-based methods reduce information redundancy and retain effective features by transforming the input image into a new feature space. Commonly used transformation-based methods include principal component analysis (PCA) [11], slow feature analysis (SFA) [12], Gram–Schmidt transform [13], and iteratively reweighted multivariate alteration detection (IR-MAD) [14]. Some commonly used CD methods, such as support vector machine (SVM) [15] and extreme learning machine (ELM) [16], belong to classification methods. However, traditional classification-based CD algorithms are more influenced by the classifier compared with deep learning (DL) methods, which limits the application of classification-based methods [17] to some extent.

Traditional methods focus primarily on generating difference maps and achieving remarkable results. However, these methods use handcrafted feature representations that limit their ability to represent complex high-level variation information.

Morphology-based methods have shown great potential in HSI-CD tasks. For example, Hou et al. [18] proposed a morphological method that extracts area and height attributes of HSI using the max-tree/min-tree strategy. To fully consider the local spatial similarity of spectra and the global topological structure of images, Hou and Li [19] introduced

Manuscript received 8 November 2022; revised 4 June 2023; accepted 4 August 2023. Date of publication 14 August 2023; date of current version 5 September 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62276197 and Grant 62171332. *(Corresponding author: Xiangrong Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3304681

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a joint morphological and patch-tensor (JMPT)-based CD method. They first mined spatial structure correlation using a patch-based tensor strategy and then extracted various morphological features to obtain different attributes of multitemporal images. These results were fused using tensor methods to obtain the final CD map.

The tensor-based approach has effectively represented high-dimensional data structures, making it a promising method for HSI-CD. Huang et al. [20] combined tensor-based and DL methods to propose a hyperspectral RS (HSRS) image CD technique. They established a tensor information model to capture latent feature changes in HSRS images and designed a three-order tensor-based restricted Boltzmann machine (Tensor3-RBM). To fully consider the intrinsic structural information of spectra, Hou et al. [21] proposed a new three-order tucker decomposition and reconstruction detector for HSI-CD. This strategy can weaken the influence of various noise signals in the dataset and validate that tensor processing can amplify the difference between unchanged and changed pixel pairs.

Recently, DL-based methods have been proposed due to their great potential performance on HSI-CD, which can mine semantic features from different levels and solve high-dimensional problems to a greater extent. For optical aerial images, Zhan et al. [22] proposed a supervised CD method based on a deep Siamese convolutional network with weighted contrast loss to cope with sample imbalance, which directly extracts features from image pairs, and the extracted features are more abstract and robust than those produced by handcrafted. It is well known that different spectral channels and spatial locations contain different amounts of information and, therefore, have different effects on the results of HSI-CD [23]. To this end, Wang et al. [23] proposed a spectral–spatialwise attention-based Siamese network (SSA-SiamNet), which adaptively enhances spectral–spatial features by emphasizing information-rich channels and locations. Although DL approaches have made outstanding contributions to HSI-CD, there are still several issues that need to be addressed.

1) HSI contains approximately continuous spectral information and can be utilized to identify changes by capturing subtle spectral differences. However, convolutional neural network (CNN) can extract local spectral sequence features, but cannot well extract and represent the global spectral features, due to the limitation of its inherent network backbone.

2) Due to the change of imaging conditions and environment, objects with the same semantic concept at the same spatial location may exhibit different spectral features at different times. Using only spectral features may misjudge the "pseudo-changes" caused by environmental changes. Previous methods have typically extracted semantic features separately from images at different times in the encoder and decoder stages without considering the temporal correlations between features. This approach is limited in its ability to fully extract and effectively utilize temporal–spatial–spectral features.

3) During the process of extracting spatial features, a simple fusion of nearly continuous spectral bands can disrupt the sequential properties of spectral features. This approach may result in low efficiency in utilizing spectral information and can greatly increase the difficulty of mining spectral sequential properties.

To address the above problems, we draw inspiration from the global modeling ability of transformers and propose a novel approach called cascade spectral-aware transformer (CAST) for the HSI-CD network, consisting of a temporal–spatial transformer (TS-Former), a spectral-awareness transformer (SA-Former), and a spectral constraint module (SCM). Specifically, we first design a TS-Former, which extracts temporal–spatial features to identify potential change regions. By doing so, we reduce the misjudgment of "pseudo-change" regions caused by objects with the same semantic concept but differing spectral features. We then propose an SA-Former to enhance the spectral features. This component has a powerful medium- and long-term modeling capability that fully exploits and represents the sequence properties of spectral features. Finally, we introduce the SCM to better maintain the sequence properties of the HSI spectrum while extracting temporal–spatial features. The SCM utilizes spectral sequence constraints to achieve better preservation of the spectral sequence properties, which ultimately leads to improved accuracy of HSI-CD.

The main contributions of this article are summarized as follows.

1) We propose a transformer architecture for HSI-CD, called CAST, which achieves outstanding performance by fully mining and effectively utilizing temporal–spatial–spectral features. Moreover, we qualitatively and quantitatively verify the effectiveness and robustness of our method on China, River, and Hermiston datasets and achieve state-of-the-art results.

2) To address the insufficient consideration of temporal correlation, we propose a TS-Former to reinforce the temporal correlation and spatial global relationship of extracted features.

3) To better mine and represent the sequence properties of spectral features, especially for medium- and long-term dependencies, we design an SA-Former that enhances the network's ability to model global spectral relationships.

4) We also observe that the process of extracting temporal–spatial features often causes spectral distortion. To alleviate this, we introduce an SCM that further enhances the sequence properties of spectral features.

The rest of this article is organized as follows. Section II introduces the related work of this article. Section III introduces the proposed framework for HSI-CD in detail. Section IV introduces the dataset and evaluation metrics. Section V presents and analyzes the experimental results. Finally, the conclusion of this article is drawn in Section VI.

II. Related Work

A. DL-Based HSI-CD Method

DL methods have powerful feature extraction and learning capabilities compared with traditional methods and have achieved excellent performance in various computer vision tasks. In the field of HSI-CD, various DL methods have

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been proposed. Lin et al. [24] proposed a bilinear CNNs (BCNNs) that effectively capture the relationship between images at different times by performing linear outer product operations, resulting in the extraction of change information. Zhan et al. [25] combined 1-D convolution and 2-D convolution and proposed a three-direction spectral–spatial convolution neural network (TDSSC) for HSI-CD, enabling the extraction of spectral information and spectral–spatial information along the spectral direction and two spatial directions, respectively. Wang et al. [26] presented a general end-to-end 2-D convolutional neural network (GETNET) framework, which makes full use of the abundance information of endmembers at the subpixel level by unmixing endmembers. They also introduced a mixed-affinity matrix that provides richer cross-channel gradient information. However, most current methods extract spectral features and spatial features separately, which is difficult the detection of "pseudo-change" in images. A 3-D convolutional autoencoder (3DCAE) method was proposed by Zhao et al. [27], which extracts spatial–spectral features using a simplified 3-D convolutional autoencoder (S3DCAECD) without prior knowledge, effectively reducing the spectral redundancy and overfitting. To solve the problems of missing small objects and rough edges in the results, Zhang et al. [28] proposed a Siamese multiscale difference-enhancement network (SMD-Net) for RS image CD.

Since HSI-CD is to solve the problem of similarities and differences between multitemporal images, the temporal relationships between images need to be considered, so recurrent convolutional neural networks (ReCNNs) were introduced. Mou et al. [29] combined CNNs and RNNs to propose an ReCNN for multispectral images, using CNNs to extract rich spectral–spatial features and then using RNNs adaptively to learn the temporal correlation between multitemporal images. Yang et al. [30] have introduced a novel deep multiscale pyramidal network with enhanced spatial–spectral residual attention (DMPs^{2}raN) for HSI-CD. This approach incorporates a convolutional LSTM (ConvLSTM) network to explore temporal correlations between images by leveraging the extracted features from T_1 and T_2 moments. DMPs^{2}raN exhibits remarkable performance in complex changing regions because of its powerful multiscale spatial–spectral feature mining capability. Qu et al. [31] proposed a multilevel encoder–decoder attention network (ML-EDAN), which utilizes a multilevel fusion strategy across multiple phases to fully utilize and fuse the hierarchical features of all convolutional layers, and a contextual-information-guided attention (CIGA) module was developed in order to transfer low-level features to higher levels efficiently. After obtaining the multilevel features, a long short-term memory (LSTM) network analyzes the temporal correlation between the multitemporal images. Graph convolution can capture the feature and structure information of non-Euclidean data through convolution operations in non-Euclidean space. Zhang et al. [32] proposed a spatial pooling graph convolutional network (SPGCN) for HSI classification, and the graph pooling operation can preserve classification-specific features and reduce redundant pixels. The dual-branch difference amplification graph convolutional network (D^2AGCN) was proposed by Qu et al. [33]. The differential amplification module in the network can suppress invariant features to highlight changing features, thus improving the accuracy of HSI-CD.

Unlike the supervised approach, the unsupervised approach does not require much labeled data. Li et al. [34] proposed a new unsupervised framework based on noise modeling for HSI-CD, where the CD maps generated by the unsupervised approach are used as pseudo-labels to train CNN. The noise is modeled in the end-to-end process with the aim of removing noise interference. Unsupervised methods can generate pseudo-labels in a model-driven manner, but the model-driven approach limits the final detection accuracy. Li et al. [35] combined two complementary model-driven approaches, structural similarity (SSIM) and CVA, to generate plausible pseudo-labels.

B. Transformer-Based Model

Although CNN has achieved good results in various computer vision tasks, it has yet to have the ability to model globally. Recently, transformer-based deep models have performed very strongly in various fields. The transformer was first proposed in [36], initially used as a sequence-to-sequence machine translation model. Transformers are now widely used in various fields, such as natural language processing (NLP), computer vision, and audio processing. Transformer has recently demonstrated powerful representation capabilities in various computer vision tasks, including image classification [37], [38], [39], video classification [40], video recognition [41], [42], [43], image segmentation [44], [45], image caption [46], [47], and image object detection [48], [49], [50].

Recently, researchers have also done some work on RS image CD tasks using a transformer. Bandara and Patel [51] proposed a transformer-based Siamese network that utilizes a hierarchical transformer encoder and a multilayer perceptron (MLP) decoder to efficiently extract the multiscale long-range detailed information. Chen et al. [52] proposed a bitemporal image transformer (BIT) to model the spatial–temporal contexts, using a compact set of tokens to represent high-level concepts of changes of interest in bitemporal images and exploiting the transformer in spatial–temporal tokens to associate semantic concepts. Wang et al. [53] used a transformer to extract spatial and spectral features, which improved CD accuracy but introduced many parameters. Dong et al. [54] proposed an abundance matrix correlation analysis network (AMCAN) based on hierarchical multihead self-cross-hybrid attention for HSI-CD, which highlights the subpixel-level correlation difference information in a hierarchical manner to detect subtle changes. Different from these methods, we design a more efficient and lightweight transformer architecture to improve the network's performance by modeling the global spectral sequence properties.

III. Proposed Method

This article proposes a CAST HSI-CD network, as shown in Fig. 1. It consists of three parts: a TS-Former, an SA-Former, and an SCM.

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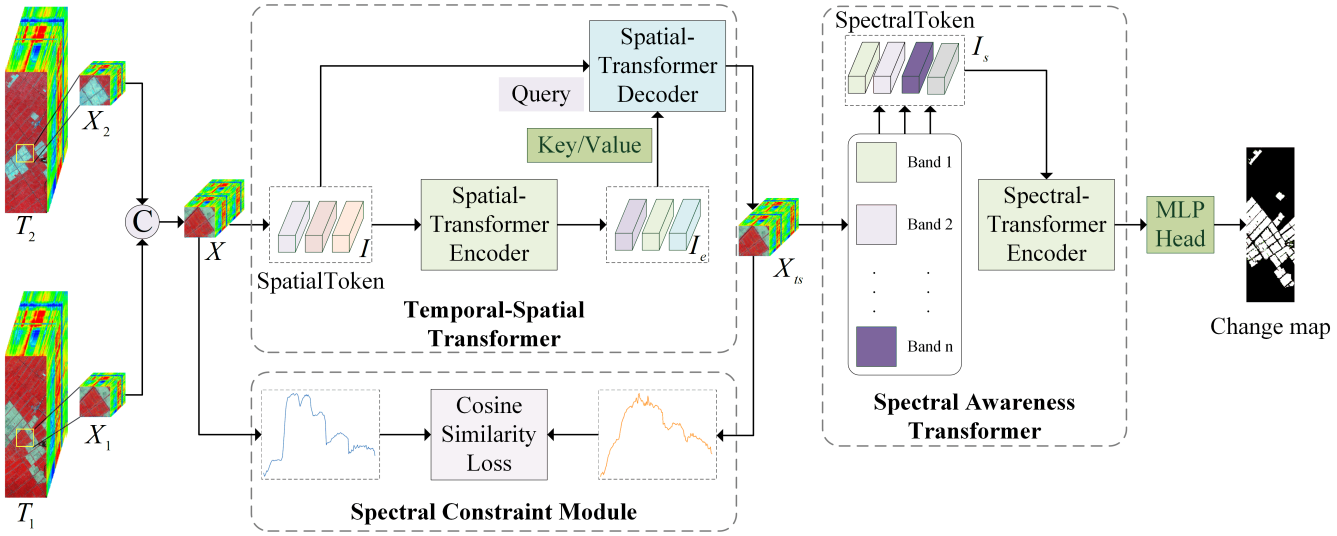


Fig. 1. Network architecture of the proposed CAST. CAST consists of three parts: the TS-Former is used to extract temporal–spatial features. The SA-Former is used to better mine and represent sequence attributes of spectral features. The SCM compensates the spectral information to reduce spectral distortion during the extraction of temporal–spatial features.

A. Overall Pipeline

The HSI-CD task involves two HSIs at the same location at different times. For the two HSIs T_1 and T_2, patches $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ of the same size are cropped from the same position of T_1 and T_2, respectively. To better learn the temporal correlation between two images, the input data $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ concatenate operation to obtain $\boldsymbol{X}$. Subsequently, $\boldsymbol{X}$ is input to the TS-Former network to extract the temporal–spatial features. The temporal–spatial features extracted by the TS-Former can be formulated as follows:

$$\boldsymbol{X} = \text{Concat}(\boldsymbol{X}_1, \boldsymbol{X}_2) \tag{1}$$

$$\boldsymbol{X}_{\text{ts}} = \text{Former}_{\text{ts}}(\boldsymbol{X}) \tag{2}$$

where $\boldsymbol{X}_{\text{ts}}$ represents temporal–spatial features extracted by the TS-Former, $\boldsymbol{X}_t$ ($t = 1$ and 2) denotes the input patch, Concat$(\cdot)$ denotes a concatenate operation along the channel dimension, and $\text{Former}_{\text{ts}}(\cdot)$ indicates a TS-Former encoder–decoder operation.

HSI contains rich spectral information. For the temporal–spatial features extracted by the TS-Former network, the SA-Former can better mine and represent the sequence attributes of spectral features, thereby obtaining temporal–spatial–spectral features. The output of the SA-Former can be formulated as follows:

$$\boldsymbol{X}_{\text{tss}} = \text{Former}_{\text{sa}}(\boldsymbol{X}_{\text{ts}}) \tag{3}$$

where $\boldsymbol{X}_{\text{tss}}$ denotes the temporal–spatial–spectral features extracted by the SA-Former and $\text{Former}_{\text{sa}}(\cdot)$ represents the spectral transformer encoder operation.

In the process of extracting temporal–spatial features, too much attention to spatial features will cause spectral distortion. The SCM introduction can better maintain spectral feature sequence properties.

HSI-CD is equivalent to a binary classification of the extracted temporal–spatial–spectral features, and classification is performed through the MLP layer to generate the final change result map. The output of the resulting map can be defined as follows:

$$\boldsymbol{F}_{\text{out}} = \text{MLP}(\boldsymbol{X}_{\text{tss}}). \tag{4}$$

B. Temporal–Spatial Transformer

Considering the temporal correlation of HSI, we first perform the concatenate operation on the input data to obtain the stack data $\boldsymbol{X} \in R^{H\times W\times 2C}$, where H and W are the length and width of the patch, respectively, and C is the number of bands of the HSI. To reduce the misjudgment of objects with the same semantic concept but different spectra due to environmental changes, the data $\boldsymbol{X}$ are first input into the TS-Former network to obtain temporal–spatial features $\boldsymbol{X}_{\text{ts}}$.

The TS-Former network consists of the TS-Former encoder and the TS-Former decoder. As shown in Fig. 2, the TS-Former encoder consists of one-layer multihead self-attention (MSA) and MLP modules. Different from vision transformer (ViT) [55], which performs layer normalization before the MSA or MLP operation and the residual connection at the end, the TS-Former encoder performs MSA or MLP operation first, followed by residual operation and last layer normalization.

1) MSA Layer: The input data $\boldsymbol{X}$ are expanded along the spatial dimension to obtain the spatial token $\boldsymbol{I} \in R^{HW\times 2C}$. Since MSA is insensitive to position information, in order to make the output of self-attention position-related, the information related to location is embedded for the input $\boldsymbol{I}$:

$$\boldsymbol{I} = \boldsymbol{I} + \boldsymbol{E}_{\text{pos}} \tag{5}$$

where $\boldsymbol{E}_{\text{pos}} \in R^{HW\times 2C}$ indicates positional embedding.

The input self-attention in the TS-Former encoder is a triple containing query $\boldsymbol{Q}$, key $\boldsymbol{K}$, and value $\boldsymbol{V}$. The calculation of $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ is as follows:

$$\begin{aligned} \boldsymbol{Q} &= \text{IW}^q \\ \boldsymbol{K} &= \text{IW}^k \\ \boldsymbol{V} &= \text{IW}^v \end{aligned} \tag{6}$$

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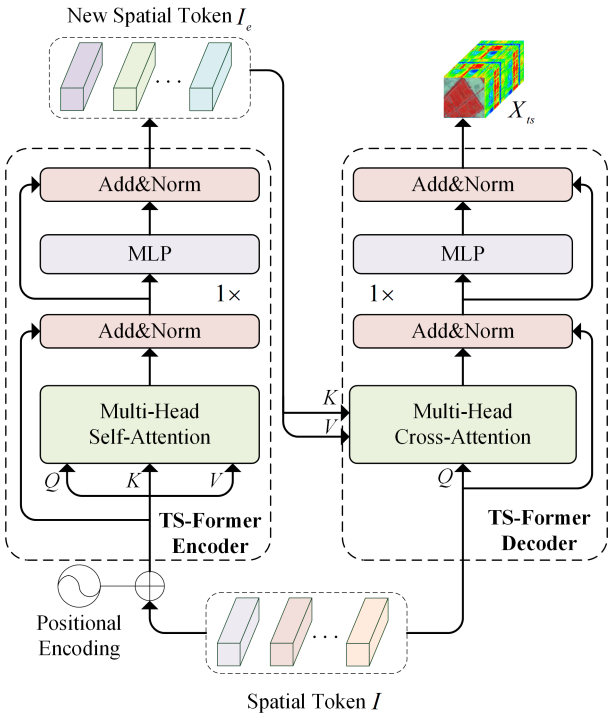


Fig. 2. Illustration of our TS-Former model. The TS-Former model includes the TS-Former encoder and the TS-Former decoder, which can fully extract temporal–spatial features.

where $\boldsymbol{W}^q, \boldsymbol{W}^k, \boldsymbol{W}^v \in R^{2C\times d_k}$ are the learnable parameters of three linear projection layers and d_k is the channel dimension of the triple. One attention head is formulated as follows:

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_k}}\right)\boldsymbol{V}. \tag{7}$$

The TS-Former encoder uses MSA, which helps the network to capture richer features and information. The equation for MSA can be defined as follows:

$$\begin{aligned}\text{MSA}(\boldsymbol{I}) &= \text{Concat}(\text{head}_1, \cdots, \text{head}_h)\boldsymbol{W}^o \\ \text{head}_j &= \text{Attention}\left(\boldsymbol{I}\boldsymbol{W}_j^q, \boldsymbol{I}\boldsymbol{W}_j^k, \boldsymbol{I}\boldsymbol{W}_j^v\right)\end{aligned} \tag{8}$$

where $\boldsymbol{W}_j^q, \boldsymbol{W}_j^k, \boldsymbol{W}_j^v \in R^{2C\times d_k}$ and $\boldsymbol{W}^O \in R^{hd_k\times 2C}$ are the linear projection matrices, h is the number of self-attention heads, and MSA($\boldsymbol{I}$) is the output of MSA.

To address the problem of network degradation and vanishing gradients, we perform residual connections on the output of MSA, which can increase the complexity of the model. However, residual connections will further expand the variance. In order to reduce the variance and speed up network training, layer normalization is introduced. The formulas for residual connection and layer normalization are as follows:

$$\boldsymbol{A}_{\text{out}} = \text{LayerNorm}(\boldsymbol{I} + \text{MSA}(\boldsymbol{I})). \tag{9}$$

2) MLP Layer: The MLP layer in the TS-Former encoder consists of two linear transforms and a Gaussian error linear unit (GELU) [56]. The MLP is formulated as follows:

$$\boldsymbol{I}_{\text{hidden}} = \text{Linear}(\text{GELU}(\text{Linear}(\boldsymbol{A}_{\text{out}}))) \tag{10}$$

$$\boldsymbol{I}_e = \text{LayerNorm}(\boldsymbol{A}_{\text{out}} + \boldsymbol{I}_{\text{hidden}}) \tag{11}$$

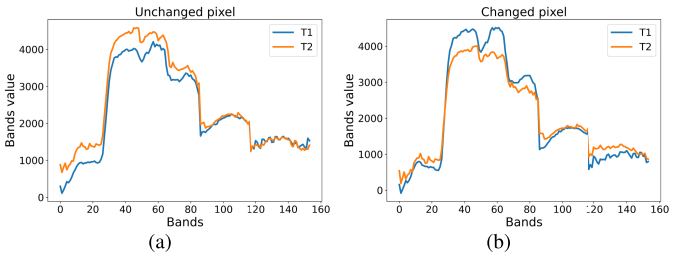


Fig. 3. Spectral waveform plot of the corresponding pixels of the China dataset. (a) Waveform plot of the bitemporal image for unchanged pixels. (b) Waveform plot of the bitemporal image for changed pixels.

where $\boldsymbol{I}_e \in R^{HW\times 2C}$ denotes the output of the TS-Former encoder.

The rich context token $\boldsymbol{I}_e$ is obtained through the TS-Former encoder, and in order to map $\boldsymbol{I}_e$ to pixel space, we use the TS-Former decoder to obtain pixel-level temporal–spatial features $\boldsymbol{X}_{\text{ts}}$. As shown in Fig. 2, our TS-Former decoder consists of one layer of multihead cross attention (MCA) and MLP. Except for MCA, all operations of the TS-Former decoder are the same as the TS-Former encoder. In MSA, the query, key, and value all come from the same input sequence $\boldsymbol{I}$, while in MCA, the query comes from $\boldsymbol{I}$, and the key and value come from the output $\boldsymbol{I}_e$ of the TS-Former encoder. MCA is defined as follows:

$$\begin{aligned}\text{MCA}(\boldsymbol{I}, \boldsymbol{I}_e) &= \text{Concat}(\text{head}_1, \cdots, \text{head}_h)\boldsymbol{W}^o \\ \text{head}_j &= \text{Attention}\left(\boldsymbol{I}\boldsymbol{W}_j^q, \boldsymbol{I}_e\boldsymbol{W}_j^k, \boldsymbol{I}_e\boldsymbol{W}_j^v\right)\end{aligned} \tag{12}$$

$$\text{MCA}_{\text{out}} = \text{LayerNorm}(\text{MCA}(\boldsymbol{I}, \boldsymbol{I}_e) + \boldsymbol{I}) \tag{13}$$

$$\boldsymbol{X}_{\text{ts}} = \text{reshape}(\text{LayerNorm}(\text{MLP}(\text{MCA}_{\text{out}}) + \text{MCA}_{\text{out}})) \tag{14}$$

where $\boldsymbol{W}_j^q, \boldsymbol{W}_j^k, \boldsymbol{W}_j^v \in R^{2C\times d_k}$ and $\boldsymbol{W}^O \in R^{hd_k\times 2C}$ are the linear projection matrices, and h is the number of attention heads. $\boldsymbol{X}_{\text{ts}}$ denotes temporal–spatial features.

C. Spectral-Awareness Transformer

Compared with optical images, HSI has near-continuous spectral information, in which the sequence properties of the spectrum can provide the fine features required for HSI-CD. However, a large number of spectral bands can also cause information redundancy. As shown in Fig. 3, the China dataset shows differences in spectral information at different times for the same location. There are still differences in the local spectral information of unchanged pixels, and modeling only local spectral information is prone to misjudgment. The changed pixels have similar spectral information in some bands, and mining the global spectral sequence properties can reduce the occurrence of false negatives (FN). Therefore, fully mining and effectively using the sequence properties of the spectrum, especially the medium- and long-term dependencies, are crucial to improve the accuracy of HSI-CD.

As shown in Fig. 4, the SA-Former is proposed to learn the sequence properties and global correlations between spectra. SA-Former can capture subtle spectral differences using a global attention mechanism to identify change regions accurately. Specifically, $\boldsymbol{X}_{\text{ts}}$ is first expanded into spectral tokens

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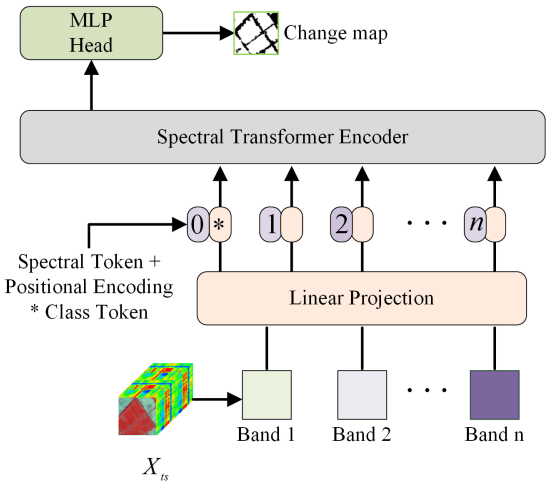


Fig. 4. Illustration of our SA-Former, the SA-Former can fully mine and effectively exploit the sequence properties of spectrum, especially the medium and long-term dependencies.

$\boldsymbol{I}_s \in R^{2C\times HW}$ according to the bands. Second, the learnable classification token ($\boldsymbol{I}_{\text{cls}} \in R^{1\times 2C}$) is embedded into the spectral token sequence, and the output of $\boldsymbol{I}_{\text{cls}}$ after SA-Former network is $\boldsymbol{I}_{\text{class}} \in R^{1\times 2C}$. Since MSA is not sensitive to position information, in order to make the output of MSA related to the band position, we introduce the location encoding $\boldsymbol{E}_{\text{pos}}$. The formula is as follows:

$$\boldsymbol{I}_s = \text{Concat}(\boldsymbol{I}_s, \boldsymbol{I}_{\text{cls}}) + \boldsymbol{E}_{\text{pos}} \tag{15}$$

where $\boldsymbol{E}_{\text{pos}} \in R^{(2C+1)\times HW}$ indicates positional embedding.

The SA-Former consists of a layer of MSA and MLP. MSA can better model the global relationship between spectra and fully use spectral features' sequence properties. The calculation process of MSA and MLP is as follows:

$$\boldsymbol{I}_{\text{hidden}} = \text{LayerNorm}(\text{MSA}(\boldsymbol{I}_s) + \boldsymbol{I}_s) \tag{16}$$

$$\boldsymbol{I}_{\text{tss}} = \text{LayerNorm}(\text{MLP}(\boldsymbol{I}_{\text{hidden}}) + \boldsymbol{I}_{\text{hidden}}) \tag{17}$$

$$y = \text{MLP}(\boldsymbol{I}_{\text{class}}) \tag{18}$$

where $\boldsymbol{I}_{\text{tss}}$ represents the output of the SA-Former network and y denotes the CD result.

With the addition of SA-Former, the sequence properties of the spectra can be fully explored and effectively utilized, especially the medium- and long-term dependences, which enhances the accuracy of HSI-CD.

D. Spectral Constraint Module

HSI contains rich spectral information, and the sequence properties of nearly continuous spectra are very effective for detecting changing objects with subtle differences in the spectrum. We discover that spatial feature extraction often causes distortion of spectral information in the process of extracting temporal–spatial features. This distortion will increase the difficulty of mining the diagnostic spectral sequence properties.

To better maintain the sequence properties of the spectrum, we propose the SCM, which can compensate for the spectral information and reduce the distortion of the spectrum. Specifically, the SCM calculates the cosine of the angle between $\boldsymbol{X}$ and $\boldsymbol{X}_{\text{ts}}$ through the cosine similarity function, which measures the degree of similarity between them. The smaller the distortion of $\boldsymbol{X}$ and $\boldsymbol{X}_{\text{ts}}$, the larger the cosine value, in the range [0,1]. To measure the distortion degree of spectral features, we use the loss L_{SCM}, the smaller the L_{SCM} value, the better the preservation of the sequence properties of the spectrum. L_{SCM} is defined as follows:

$$\begin{aligned} L_{\text{SCM}}(\boldsymbol{X}, \boldsymbol{X}_{\text{ts}}) &= 1 - \cos(\boldsymbol{X}, \boldsymbol{X}_{\text{ts}}) \\ &= 1 - \frac{1}{Q}\sum_{k=1}^{Q} \frac{\boldsymbol{X}^{(k)} \cdot \boldsymbol{X}_{\text{ts}}^{(k)}}{\left\|\boldsymbol{X}^{(k)}\right\|_2 \left\|\boldsymbol{X}_{\text{ts}}^{(k)}\right\|_2} \end{aligned} \tag{19}$$

where $\boldsymbol{X}^{(k)}$ denotes the spectral vector of the kth pixel of $\boldsymbol{X}$, $\boldsymbol{X}_{\text{ts}}^{(k)}$ denotes the spectral vector of the kth pixel of $\boldsymbol{X}_{\text{ts}}$ ($1 \le k \le Q$), and Q is the number of patch pixels. $\cos(\boldsymbol{X}, \boldsymbol{X}_{\text{ts}})$ represents the cosine of the angle between $\boldsymbol{X}$ and $\boldsymbol{X}_{\text{ts}}$.

E. Loss Function

For the optimization of the proposed CAST model, it is crucial to design an objective function reasonably. HSI-CD needs to determine if every pixel has changed, which can be regarded as a binary classification problem. Therefore, the cross-entropy loss is used to measure the classification error. The binary cross entropy can be expressed as follows:

$$L_{\text{BCE}} = -\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log \hat{y}_i + (1-y_i)\log\left(1-\hat{y}_i\right)\right) \tag{20}$$

where n denotes the number of samples, y_i is the label of ground truth whose value is 0 or 1, and $\hat{y}_i$ represents the predicted probability value, which is calculated by the softmax function.

We propose a new loss function that combines spectral constraint loss and cross-entropy loss with weighing to reduce spectral distortion and classification error simultaneously. Specifically, the overall loss function can be defined as follows:

$$\begin{aligned} L_{\text{overall}} = \lambda_1 &\left(-\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log \hat{y}_i + (1-y_i)\log\left(1-\hat{y}_i\right)\right)\right) \\ &+ \lambda_2 \sum_{i=1}^{n}\left(1 - \frac{1}{Q}\sum_{k=1}^{Q}\frac{\boldsymbol{X}^{(k)} \cdot \boldsymbol{X}_{ts}^{(k)}}{\left\|\boldsymbol{X}^{(k)}\right\|_2 \left\|\boldsymbol{X}_{ts}^{(k)}\right\|_2}\right) \end{aligned} \tag{21}$$

where λ_1 and λ_2 represent weighting coefficients. $\boldsymbol{X}^{(k)}$ denotes the spectral vector of the kth pixel of $\boldsymbol{X}$, and $\boldsymbol{X}_{\text{ts}}^{(k)}$ denotes the spectral vector of the kth pixel of $\boldsymbol{X}_{\text{ts}}$ ($1 \le k \le Q$). Q is the number of patch pixels.

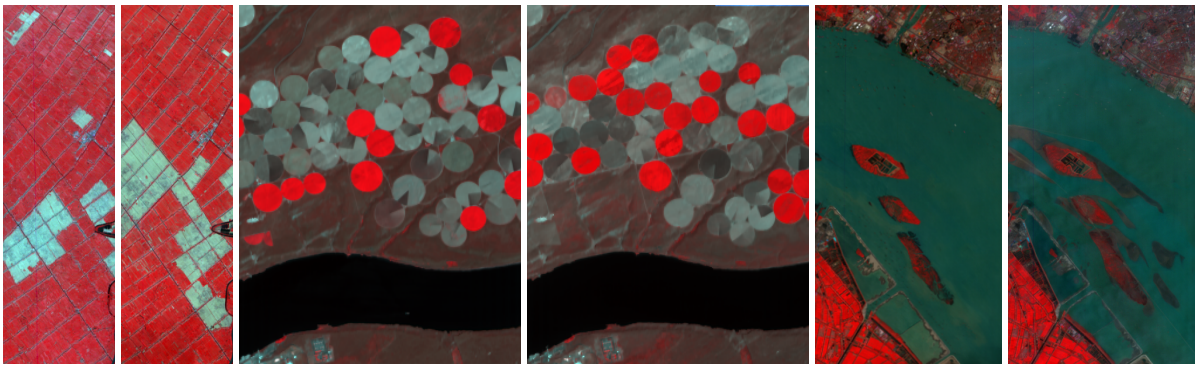
IV. Datasets and Evaluation Metrics

A. Datasets

To demonstrate the effectiveness of the CAST method, we validate it on three publicly available HSI datasets. The China [26], River [26], and Hermiston[1] datasets were acquired by the Hyperion sensor on the EO-1 satellite. The HSIs

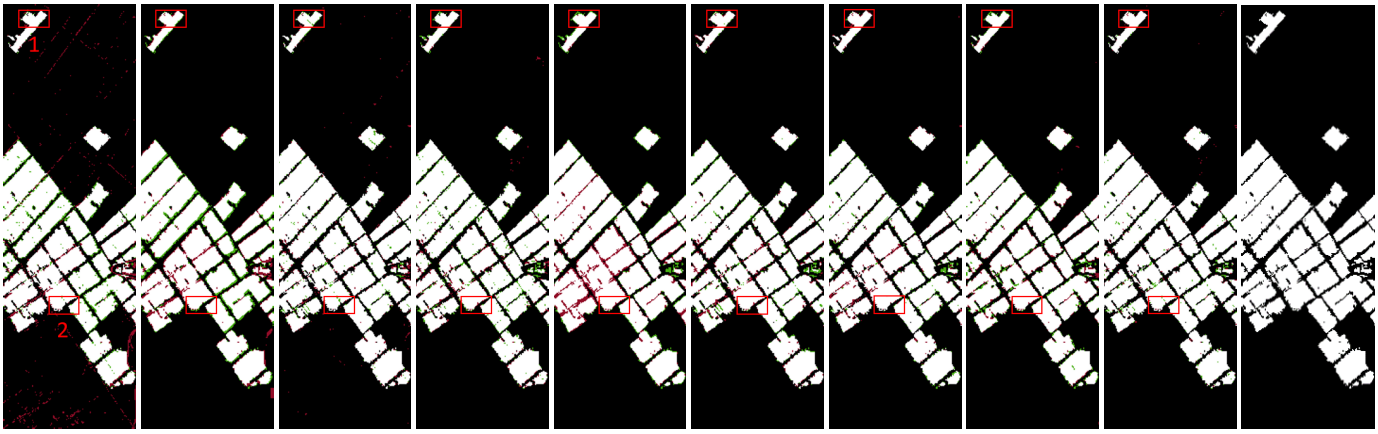
[1] https://rslab.ut.ac.ir/data

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(a) (b) (c) (d) (e) (f)

Fig. 5. Pseudo-color images of China, Hermiston, and River datasets acquired by EO-1 Hyperion sensor. (a) China image acquired on May 3, 2006. (b) China image acquired on April 23, 2007. (c) Hermiston image acquired on May 1, 2004. (d) Hermiston image acquired on May 8, 2007. (e) River image acquired on May 3, 2013. (f) River image acquired on December 31, 2013.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 6. Visualization results of different methods on China Dataset. (a) CVA. (b) PCA. (c) SVM. (d) ReCNN. (e) BCNNs. (f) ML-EDAN. (g) SST-Former. (h) AMCAN. (i) CAST. (j) Ground-truth map. White for TP, black for TN, red for FP, and green for FN.

acquired by the EO-1 Hyperion sensor have a spatial resolution of 30 m and a spectral resolution of 10 nm. It contains 242 spectral bands in the spectral range of 0.4–2.5 μm. Fig. 5 shows the images of the three-channel pseudo-color composite (bands 39, 19, and 9 as RGB).

The China dataset covers farmland near Yancheng, Jiangsu, China, as shown in Fig. 5(a) and (b). The bitemporal images were acquired on May 3, 2006 and April 23, 2007. The size of each image is 450 $\times$ 140 pixels, and 155 bands are used for CD after removing the atmospheric water absorption bands. Fig. 6(j) is the ground truth of the China dataset, which contains 44 723 changed pixels and 18 277 unchanged pixels.

As shown in Fig. 5(c) and (d), the Hermiston dataset belongs to an irrigated agricultural field of Hermiston, Umatilla, OR, USA, which was acquired on May 1, 2004 and May 8, 2007, respectively. The size of each image is 307 $\times$ 241 pixels, including 57 311 unchanged pixels and 16 676 changed pixels. After removing the water absorption bands in the atmosphere, 154 bands are selected for the experiments. Changes are mainly related to the type of land cover and river.

As shown in Fig. 5(e) and (f), the River [26] dataset mainly records the material changes of a river in Jiangsu, China, acquired on May 3, 2013 and December 31, 2013, respectively. The image size is 463 $\times$ 241 pixels and contains 101 885 unchanged pixels and 9698 changed pixels. After removing the water absorption bands in the atmosphere, 198 bands are selected in the experiment.

TABLE I
CONFUSION MATRIX

Confusion Matrix		Predicted Change	Predicted Unchange
Actual	Change	TP	FN
	Unchange	FP	TN

The overall accuracy (OA) and kappa coefficient (KC) [57] are used to quantitatively evaluate the performance of the CAST method. OA can measure the ratio of correctly classified samples to the total samples. A larger OA value indicates a better CD result obtained. According to Table I, OA is calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{22}$$

KC can assess the consistency of the CD result map and the ground-truth map. Compared with OA, KC can more objectively reflect the accuracy of the results. The higher the value of KC, the better the CD result, and KC is calculated

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TABLE II
HYPERPARAMETERS OF CAST AND COMPARATIVE EXPERIMENTAL METHODS

method	dim	heads	blocks	patch size	batch size	optim	epoch	lr
ReCNN [29]	3	-	2	7	64	Adam	800	0.0002
BCNNs [24]	3	-	3	7	64	Adam	800	0.0002
ML-EDAN [31]	3	-	6	5	16	Adam	200	0.0001
SST-Former [53]	32/256	4	8	5	64	Adam	400	0.0005
AMCAN [54]	256	4	2	5	16	Adam	200	0.0001
CAST	310/49	10	3	7	128	Adam	200	1



Fig. 7. Two enlarged sub-regions with different methods on the China Dataset. (a) CVA. (b) PCA. (c) SVM. (d) ReCNN. (e) BCNNs. (f) ML-EDAN. (g) SST-Former. (h) AMCAN. (i) CAST. (j) Ground-truth map. White for TP, black for TN, red for FP, and green for FN.

as follows:

$$\mathrm{KC} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{23}$$

$$\mathrm{PRE} = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} + \frac{(\mathrm{FN}+\mathrm{TN})(\mathrm{FP}+\mathrm{TN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2}. \tag{24}$$

V. EXPERIMENT AND ANALYSIS

A. Compared Methods and Experimental Setup

To verify the effectiveness of our proposed method, eight state-of-the-art algorithms are selected for comparison, such as CVA [9], PCA [11], SVM [15], ReCNN [29], BCNNs [24], ML-EDAN [31], spectral, spatial, and temporal transformer (SST-Former) [53], and AMCAN [54]. The hyperparameters for various comparative experiments are shown in Table II. CVA and PCA are unsupervised methods, while SVM, ReCNN, and BCNNs are supervised methods. CVA is a classical algebra-based CD method that first calculates the Euclidean distance for all pixels to generate a difference map and then clusters the difference map to obtain the final result. PCA preserves key information by selecting representative features and then performs K-means clustering on the extracted key feature vectors to obtain the results of CD. SVM is a classical classification-based supervised method that uses radial basis function (RBF) kernel computation to generate a high-dimensional hyperplane to distinguish image-changed and unchanged pixels. Among the DL-based methods, ReCNN, BCNNs, ML-EDAN, SST-Former, and AMCAN have been selected as comparative methods, because they have shown excellent performance in HSI-CD.

The proposed CAST is trained on an NVIDIA GTX 2080Ti with 11G GPU through the PyTorch framework. The hyperparameters are shown in Table II. Adadelta [58] is chosen as the optimizer for the network and trained for 200 epochs. The initial learning rate of the network is set to 1, and the learning rate decreases by a factor of 10 every 80 epochs. The batch size is set to 128, and the patch size is 7×7. For the River and Hermiston datasets, the selection of the training set and the test set is consistent with [30] and [31], where the changed and unchanged pixels are selected at a 1:1 ratio and 20% of the data are used for training, with the rest used for testing. For the China dataset, the training set and the test set selection is consistent with [23], where 5% of the data are used for training, and the rest are used for testing. To increase the number of training sets, we flip the training images vertically, horizontally, and diagonally, and the training set is expanded by a factor of 4. To mitigate the impact of random sample selection on the results in DL methods, each dataset was run four times, and the mean and standard deviation were calculated to obtain the final results.

B. Experimental Results

To qualitatively measure the effectiveness of the proposed method, we visualize the experimental results of the proposed and compare methods. As shown in Figs. 6–9, the performance of each method can be more intuitively evaluated. In addition, we quantitatively evaluate each method using objective indicators OA and KC, such as Tables III–V, and the evaluation of the indicators is more comprehensive and accurate.

1) Experiment on the China Dataset: As shown in Fig. 6, the main changes are caused by changes in the size of the farmland. Because CVA cannot choose a suitable threshold to distinguish the unchanged regions, a large number of scattered red pixels appear in Fig. 6(a). Compared with CVA, PCA has improved the detection results of unchanged regions, but there are still many misjudgments and FN for the boundaries of changed regions. Compared with unsupervised methods, SVM, ReCNN, and BCNNs improve the overall detection accuracy through training. In Fig. 6(e), many unchanged regions at the boundary of changing regions are detected as changing regions, which is because BCNNs do not fully consider the

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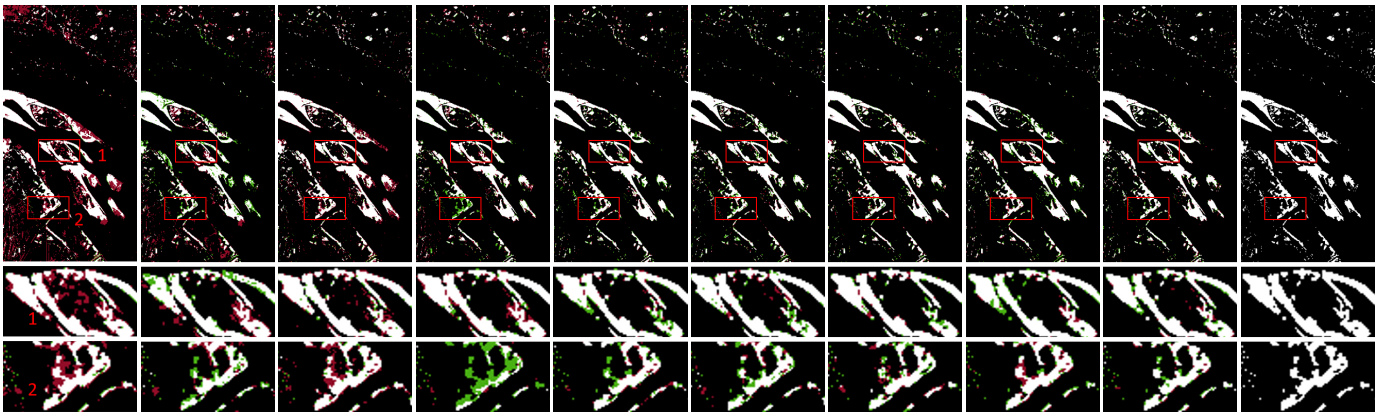


Fig. 8. Visualization results of different methods on River Dataset. (a) CVA. (b) PCA. (c) SVM. (d) ReCNN. (e) BCNNs. (f) ML-EDAN. (g) SST-Former. (h) AMCAN. (i) CAST. (j) Ground-truth map. White for TP, black for TN, red for FP, and green for FN.

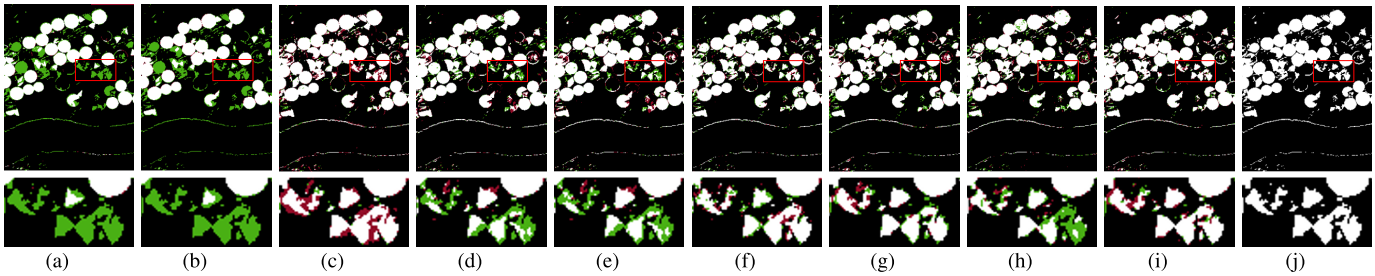


Fig. 9. Visualization results of different methods on Hermiston Dataset. (a) CVA. (b) PCA. (c) SVM. (d) ReCNN. (e) BCNNs. (f) ML-EDAN. (g) SST-Former. (h) AMCAN. (i) CAST. (j) Ground-truth map. White for TP, black for TN, red for FP, and green for FN.

TABLE III

QUANTITATIVE ASSESSMENT OF CAST AND OTHER STATE-OF-THE-ART CD METHODS ON THE CHINA DATASET

China	OA(%)	KC	TP↑	TN↑	FP↓	FN↓
CVA [9]	95.80	0.8986	17114	43240	1483	1163
PCA [11]	95.37	0.8867	16601	43479	1244	1676
SVM [15]	98.18	0.9559	16859	41904	583	505
ReCNN [29]	97.18±0.28	0.9314±0.0065	16440	41634	853	924
BCNNs [24]	96.38±0.10	0.9126±0.0026	16362	41304	1183	1002
ML-EDAN [31]	97.71±0.05	0.9444±0.0011	16632	41858	629	732
SST-Former [53]	97.86±0.10	0.9487±0.0026	17028	41581	906	336
AMCAN [54]	97.47±0.09	0.9382±0.0023	16522	41864	661	859
CAST	**98.59±0.09**	**0.9657±0.0023**	16892	42200	287	472

TABLE IV

QUANTITATIVE ASSESSMENT OF CAST AND OTHER STATE-OF-THE-ART CD METHODS ON THE RIVER DATASET

River	OA(%)	KC	TP↑	TN↑	FP↓	FN↓
CVA [9]	92.84	0.6624	9314	94275	7610	384
PCA [11]	95.67	0.7350	7602	99147	2738	2096
SVM [15]	96.65	0.8115	9127	98713	3172	571
ReCNN [29]	96.96±0.09	0.8036±0.0065	8090	100140	1745	1608
BCNNs [24]	97.94±0.22	0.8690±0.0140	8393	100956	929	1305
ML-EDAN [31]	98.08±0.07	0.8771±0.0032	8454	100988	897	1244
SST-Former [53]	97.98±0.29	0.8742±0.0135	8490	100956	929	1208
AMCAN [54]	97.95±0.12	0.8686±0.0076	8357	100942	943	1341
CAST	**98.40±0.04**	**0.8988±0.0027**	8784	101075	810	914

temporal correlation between images. Fig. 6(f)–(h), based on DL state-of-the-art methods, significantly improves the accuracy of detection. In contrast, the proposed CAST method contains fewer red pixels, indicating less misjudgment phenomenon. It can be seen from the intercepted subregions that the CAST method contains the least number of red and green pixels, indicating that the proposed method has the best performance and can reduce the occurrence of misjudgment and missed judgment.

The best results are shown in bold in Table III. The OA and KC values of the supervised method are higher than the unsupervised CVA and PCA methods. However, the results of SVM outperform the results of all comparative experiments. Relative to SST-Former, our proposed method achieves the OA and KC values of 98.59% and 0.9657, respectively, surpassing all state-of-the-art comparison methods, proving the effectiveness of our method. The values of FN and false positives (FP) are smaller, indicating that the phenomenon of missed judgment and misjudgment is also the least.

2) Experiment on the River Dataset: Fig. 8 shows the visualization results of various methods on the River dataset. Many unchanged regions in Fig. 8(a) are misjudged as changed regions by the CVA method. In Fig. 8(b), although the red area has decreased, the PCA has more seriously missed judgment

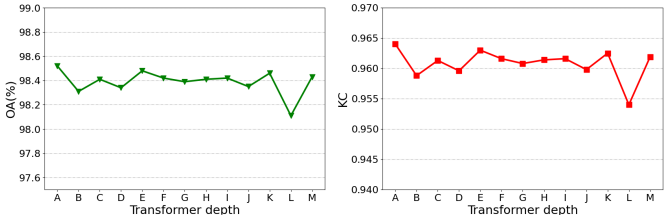
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TABLE V
QUANTITATIVE ASSESSMENT OF CAST AND OTHER STATE-OF-THE-ART CD METHODS ON THE HERMISTON DATASET

Hermiston	OA(%)	KC	TP↑	TN↑	FP↓	FN↓
CVA [9]	91.99	0.7406	11002	57056	255	5674
PCA [11]	92.14	0.7456	11073	57095	216	5603
SVM [15]	96.17	0.8932	15932	55218	2093	744
ReCNN [29]	96.09±0.04	0.8868±0.0013	14986	56108	1203	1690
BCNNs [24]	96.02±0.06	0.8845±0.0022	15151	55901	1410	1525
ML-EDAN [31]	97.07±0.06	0.9139±0.0012	15589	56131	1180	1087
SST-Former [53]	96.88±0.09	0.9095±0.0037	15300	56371	940	1376
AMCAN [54]	96.89±0.48	0.9100±0.0141	15314	56297	1014	1362
CAST	**97.33±0.24**	**0.9232±0.0065**	15804	56379	932	872

TABLE VI
COMPUTATIONAL COST OF DIFFERENT METHODS ON THE CHINA DATASET

Method	Params	OA(%)	KC
ReCNN [29]	0.65M	97.18±0.28	0.9314±0.0065
BCNNs [24]	0.82M	96.38±0.10	0.9126±0.0026
ML-EDAN [31]	112.72M	97.71±0.05	0.9444±0.0011
SST-Former [53]	2.51M	97.86±0.10	0.9487±0.0026
AMCAN [54]	75.82M	97.47±0.09	0.9382±0.0023
CAST	0.92M	**98.59±0.09**	**0.9657±0.0023**



(a) (b)

Fig. 10. Performance comparison of CAST networks with different depths on the China dataset. (a) OA values of CAST at different depths. (b) KC values of CAST at different depths. *a* represents the number of TS-Former encoders, *b* represents the number of TS-Former decoders, and *c* represents the number of SA-Former. In the above figure, from left to right (A–M), *a*–*c* are (1, 1, 1), (1, 2, 1), (1, 1, 2), (2, 1, 1), (2, 2, 1), (2, 2, 2), (1, 3, 1), (1, 1, 3), (3, 1, 1), (1, 2, 3), (2, 3, 4), (3, 3, 3), and (4, 4, 4), respectively.

for some scattered change regions. Compared with CVA and PCA, there are fewer red and green pixels in the SVM visualization result, indicating that the performance of SVM is better. In the DL method, it can be clearly seen from the two subregions that the ReCNN method has the worst antinoise interference ability, and there are many FN and misjudgments. In some recent advanced methods, BCNNs, ML-EDAN, SST-Former, and AMCAN have achieved good results, and the red and green pixels in the figure are significantly reduced. The resulting map of the proposed CAST method has the least red and green pixels, which is the closest to the ground-truth map, proving that the proposed method performs best on the River dataset.

Table IV shows the quantitative index values of various methods on the River dataset. Among the traditional methods, the OA and KC values of the CVA method are the lowest, and the SVM method is the best. Compared with the traditional method, the OA and KC values of the DL method are improved, reaching more than 96.96% and 0.8036, respectively. The variance of the four experimental results of SST-Former is 0.29% and 0.0135, and the variance of the CAST method is smaller, indicating that the performance of our proposed method is more stable. The proposed method has the highest OA and KC values, reaching 98.40% and 0.8988, which are 0.32% and 0.0216 higher than ML-EDAN, respectively. It shows that using temporal–spatial–spectral features and spectral sequence properties is beneficial to HSI-CD. FP and FN represented misjudgment and missed judgment, respectively, and the FP and FN values of the CAST method are also the lowest, indicating that our method can effectively reduce the phenomenon of misjudgment and missed judgment.

3) Experiment on the Hermiston Dataset: As shown in Fig. 9, the change of the Hermiston dataset is more complex, mainly including the coverage area of farmland and the edge of rivers. As shown in Fig. 9(a) and (b), many irregular changes are not detected by CVA and PCA methods. The performance of the SVM method has been improved, and the phenomenon of missed detection has been greatly reduced, but the phenomenon of false detection has increased. For the DL method, it can be seen from the enlarged subregion that the result graphs of the ReCNN and BCNNs methods are very similar, and many missed detections exist. It can be seen from Fig. 9(f)–(h) that there are relatively few red and green colors in the ML-EDAN and SST-Former images. Some recent classic DL methods have also achieved good results. Fig. 9(i) contains the fewest red and green pixels, demonstrating the best performance of the proposed method.

Table V lists the OA and KC values of the proposed and the comparison methods. It can be seen from Table V that the OA and KC values of the traditional methods CVA and PCA are relatively low. The OA and KC values of ReCNN and BCNNs methods are very close. ML-EDAN performs very well, the best performance of all compared methods. Compared with the transformer method SST-Former, our method improves OA and KC by 0.45% and 0.0137, respectively. Obviously, both qualitative and quantitative results show that our proposed method is optimal.

As can be seen from Tables III–VI, compared with BCNNs and ReCNN networks, the performance of our proposed method has been greatly improved with a small increase in the number of parameters, which improves the accuracy of HSI-CD. Compared with the SST-Former network of the transformer structure, the parameter quantity of our model is about 36% of SST-Former, and the performance is improved, while the parameter quantity is reduced. Compared with the advanced CNN model ML-EDAN, our parameters are about 0.82% of the ML-EDAN method, while OA and KC are improved by 0.88% and 0.0213 on the China dataset, respectively. Extensive experiments demonstrate that our proposed transformer framework is more lightweight and effective.

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TABLE VII
QUANTITATIVE EVALUATION OF SCM EFFECTIVENESS ON THE CHINA, RIVER, AND HERMISTON DATASETS

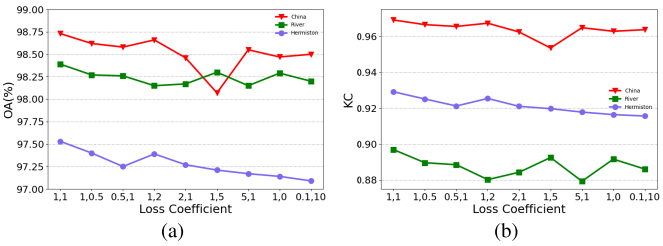
Method	SCM	China		River		Hermiston	
		OA(%)	KC	OA(%)	KC	OA(%)	KC
TS-Former	w/o-SCM	98.14±0.11	0.9550±0.0028	98.10±0.04	0.8792±0.0049	97.02±0.09	0.9143±0.0036
	w-SCM	98.16±0.27	0.9553±0.0066	98.20±0.06	0.8840±0.0048	97.26±0.15	0.9208±0.0053
SA-Former	w/o-SCM	95.88±0.08	0.9003±0.0015	97.73±0.15	0.8560±0.0092	97.12±0.04	0.9173±0.0017
	w-SCM	96.02±0.19	0.9037±0.0044	97.77±0.16	0.8587±0.0102	97.27±0.04	0.9216±0.0017
CAST	w/o-SCM	98.39±0.13	0.9609±0.0034	98.27±0.04	0.8896±0.0018	97.30±0.13	0.9220±0.0043
	w-SCM	**98.59±0.09**	**0.9657±0.0023**	**98.40±0.04**	**0.8988±0.0027**	**97.33±0.24**	**0.9232±0.0065**

C. Ablation Study

1) Effectiveness of SCM: To verify the effectiveness and stability of SCM, we conduct ablation experiments on the China, River, and Hermiston datasets, respectively. Six sets of experiments are carried out on three models, namely, TS-Former w/o-SCM (w/o-SCM means no SCM module), TS-Former w-SCM (w-SCM means using SCM module), SA-Former w/o-SCM, SA-Former w-SCM, CAST w/o-SCM, and CAST w-SCM. In Table VII, the OA and KC values of the three models using SCM on the three datasets are higher than those without SCM. The effectiveness and stability of SCM are demonstrated by quantitative evaluation results on three widely used HSI-CD public datasets. At the same time, it reflects that the sequence property of HSI can be maintained, and SCM can improve the network's performance. From Table VII, we can see that the CAST performance becomes better when combining TS-Former and SA-Former. This is because TS-Former can fully use temporal–spatial features, and SA-Former has powerful medium- and long-term modeling capabilities, which can fully mine and represent the sequence attributes of spectral features. Combining the two can make full use of the temporal–spatial–spectral features of HSIs, thereby improving the accuracy of HSI-CD.

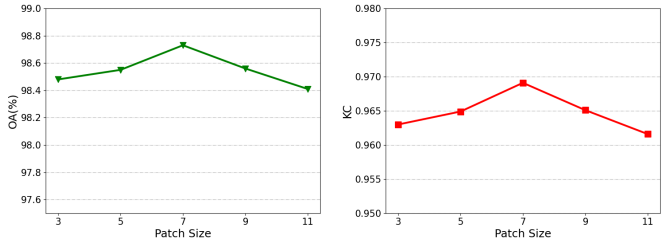
2) Depth of the Transformer: To compare the impact of transformers with different depths on network performance, we conducted a series of experiments on the China dataset. As shown in Fig. 10, the network performance fluctuates with increased network depth. Through a large number of experiments, it has been verified that when the depth of the TS-Former encoder, TS-Former decoder, and SA-Former is all set to 1, the OA and KC values of the network reach the maximum value. Therefore, in this article, we set the depth of the TS-Former encoder, TS-Former decoder, and SA-Former to 1.

3) Loss Function: The loss function of CAST is a weighted combination of cross-entropy loss and spectral constraint loss [as shown in (21)]. In (21), λ_1 and λ_2 are the weighting coefficients for the cross-entropy loss and spectral constraint loss, respectively. To verify the effectiveness of the cross-entropy loss and the spectral constraint loss, and to select the optimal coefficient combination, we use $(\lambda_1, \lambda_2) = (1, 1), (1, 0.5), (0.5, 1), (1, 2), (2, 1), (1, 5), (5, 1), (1, 0)$, and $(0.1, 10)$ to train and test the network on the China, River, and Hermiston datasets. As shown in Fig. 11, the performance of



(a) (b)

Fig. 11. Quantitative assessment results of CAST with different weighting coefficients λ_1 and λ_2 on three datasets. (a) OA values for different values of λ_1 and λ_2. (b) KC values for different values of λ_1 and λ_2.



(a) (b)

Fig. 12. Accuracy of different patch sizes on the China dataset. (a) Line graph for different patch size OA values. (b) Line graph for different patch size KC values.

different combinations of network architectures varies. When $\lambda_1 = 1$ and $\lambda_2 = 0$, the performance of the CAST network is relatively poor, which indicates the effectiveness of the proposed spectral constraint loss. When $\lambda_1 = 1$ and $\lambda_2 = 1$, the network has the highest OA and KC values on all three datasets. Therefore, through many experiments, the weighting coefficients λ_1 and λ_2 are set to 1 in this article.

4) Impact of Patch Size: The size of the patch has a certain influence on the results of the experiment. To choose the appropriate patch size, we take the patch as 3×3, 5×5, 7×7, 9×9, and 11×11, respectively, to conduct experiments on the China dataset. Fig. 12 is a visualization of the OA and KC values. It can be seen from the figure that as the patch size increases, the OA and KC values first increase and then decrease. When the patch is 7×7, the performance of the proposed network is the best. Therefore, the patch size in this article is set to 7×7.

5) Comparison With Other Attention Mechanisms: The transformer is equivalent to a global attention mechanism. To verify the effectiveness of SA-Former, we compare

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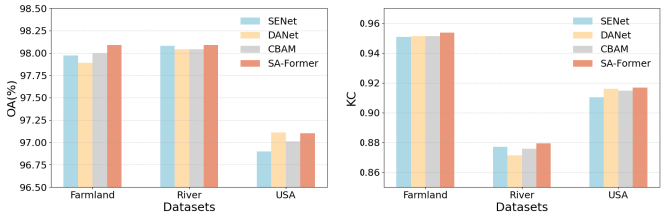


Fig. 13. Results of different attention mechanisms (from left to right: SENet, DANet, CBAM, and SA-Former) on the China, River, and Hermiston datasets. (a) Demonstration of OA values for different attention mechanisms. (b) Demonstration of KC values for different attention mechanisms.

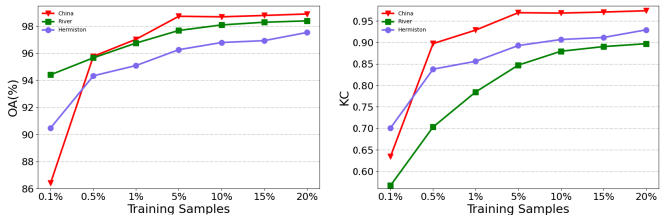


Fig. 14. Impact of different training sample proportions on the performance of the CAST model on three datasets. (a) OA values with different training sample proportions. (b) KC values with different training sample proportions.

SA-Former with other state-of-the-art attention mechanisms, including the squeeze-and-excitation network (SENet) [59], dual-attention network (DANet) [60], and convolutional block attention module (CBAM) [61]. For an objective and fair comparison, the temporal–spatial features are first extracted using the same structured CNN, and then, the extracted features are input to various attention mechanisms. Since SENet only considers spectral attention, while DANet and CBAM consider both spectral and spatial attention, DANet and CBAM outperform SENet. Fig. 13 shows that the SA-Former outperforms other attention mechanisms, fully utilizing the global spectral information can improve the accuracy of HSI-CD, proving the effectiveness of SA-Former.

6) Impact of Different Proportions of Training Samples: To investigate the impact of the training sample size on the experimental results, we conducted a series of experiments on three datasets using different sample proportions of 0.1%, 0.5%, 1%, 5%, 10%, 15%, and 20%. As illustrated in Fig. 14, there was a significant increase in both OA and KC values for all three datasets, as the training sample proportions increased from 0.1% to 5%. The China dataset's OA and KC values remained stable, as the training sample proportions increased beyond 5%. However, the model performance improved for the River and Hermiston datasets, as the training sample proportions increased from 5% to 20%. Our proposed CAST method performed better with training sample proportions of 5% for the China dataset and 20% for the River and Hermiston datasets.

7) TS-Former Attention Heatmap: We validate the ability of TS-Former to effectively capture temporal correlations in HSI-CD by visualizing its attention heatmap. Fig. 15 showcases a 70×70 subimage extracted from the top-left corner of the China dataset. In Fig. 15(a), we observe that TS-Former

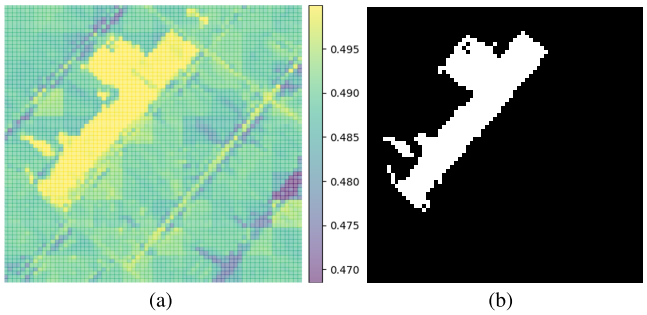


Fig. 15. Visualization of TS-Former attention heatmap on the China dataset. (a) Visualization of TS-Former attention heatmap. (b) Ground-truth map of the cropped area.

strongly focuses on the changed area after concatenating the HSI, indicating its capability to capture the temporal differences between T_1 and T_2. This experimental verification confirms that our proposed TS-Former model successfully captures the temporal correlation information between different images, further enhancing its performance in HSI-CD.

VI. Conclusion

This article proposes a pure transformer backbone for HSI-CD called CAST. CAST can fully consider images' temporal correlation and effectively utilize HSI's temporal–spatial–spectral features. Considering that HSI has approximately continuous spectral information, we model the global spectral sequence properties by SA-Former. To account for temporal correlations, we extract temporal–spatial features via TS-Former. In addition, we observe that SCM is able to maintain the properties of the original spectral sequence and reduce the occurrence of spectral distortions, thereby improving the accuracy of HSI-CD. Finally, experimental results on several public benchmarks have demonstrated the effectiveness and robustness of the CAST.

In the future, our work will be devoted to exploring an HSI semantic CD model that is able to determine the type of change, which will further facilitate the practical application of HSI-CD models.

References

[1] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.

[2] M. Gong, X. Niu, P. Zhang, and Z. Li, "Generative adversarial networks for change detection in multispectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2310–2314, Dec. 2017.

[3] F. Aslami and A. Ghorbani, "Object-based land-use/land-cover change detection using Landsat imagery: A case study of Ardabil, Namin, and Nir counties in Northwest Iran," *Environ. Monitor. Assessment*, vol. 190, no. 7, pp. 1–14, Jul. 2018.

[4] N. Camarretta et al., "Monitoring forest structure to guide adaptive management of forest restoration: A review of remote sensing approaches," *New Forests*, vol. 51, no. 4, pp. 573–596, Jul. 2020.

[5] H. Luo, C. Liu, C. Wu, and X. Guo, "Urban change detection based on Dempster–Shafer theory for multitemporal very high-resolution imagery," *Remote Sens.*, vol. 10, no. 7, p. 980, Jun. 2018.

[6] U. Bhangale, S. Durbha, A. Potnis, and R. Shinde, "Rapid earthquake damage detection using deep learning from VHR remote sensing images," in *Proc. EEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 2654–2657.

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[7] X. Zhang et al., "Spectral–spatial self-attention networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512115.

[8] D. Lu, P. Mausel, E. Brondízio, and E. Moran, "Change detection techniques," *Int. J. Remote Sens.*, vol. 25, no. 12, pp. 2365–2401, 2004.

[9] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2007.

[10] M. Hasanlou and S. T. Seydi, "Hyperspectral change detection: An experimental comparative study," *Int. J. Remote Sens.*, vol. 39, no. 20, pp. 7029–7083, Oct. 2018.

[11] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, Aug. 2008.

[12] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.

[13] J. B. Collins and C. E. Woodcock, "An assessment of several linear change detection techniques for mapping forest mortality using multitemporal Landsat TM data," *Remote Sens. Environ.*, vol. 56, no. 1, pp. 66–77, Apr. 1996.

[14] A. Aasbjerg Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

[15] H. Nemmour and Y. Chibani, "Multiple support vector machines for land cover change detection: An application for mapping urban extensions," *ISPRS J. Photogramm. Remote Sens.*, vol. 61, no. 2, pp. 125–133, Nov. 2006.

[16] G.-B. Huang, Q.-Y. Zhu, and C.-K. Siew, "Extreme learning machine: Theory and applications," *Neurocomputing*, vol. 70, nos. 1–3, pp. 489–501, Dec. 2006.

[17] C. Wu, B. Du, and L. Zhang, "A subspace-based change detection method for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 815–830, Apr. 2013.

[18] Z. Hou, W. Li, L. Li, R. Tao, and Q. Du, "Hyperspectral change detection based on multiple morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5507312.

[19] Z. Hou and W. Li, "A joint morphological profiles and patch tensor change detection for hyperspectral imagery," 2022, *arXiv:2201.08027*.

[20] F. Huang, Y. Yu, and T. Feng, "Hyperspectral remote sensing image change detection based on tensor and deep learning," *J. Vis. Commun. Image Represent.*, vol. 58, pp. 233–244, Jan. 2019.

[21] Z. Hou, W. Li, R. Tao, and Q. Du, "Three-order tucker decomposition and reconstruction detector for unsupervised hyperspectral change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6194–6205, 2021.

[22] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.

[23] L. Wang, L. Wang, Q. Wang, and P. M. Atkinson, "SSA-SiamNet: Spectral–spatial-wise attention-based Siamese network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022, Art. no. 5510018.

[24] Y. Lin, S. Li, L. Fang, and P. Ghamisi, "Multispectral change detection with bilinear convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1757–1761, Oct. 2020.

[25] T. Zhan et al., "TDSSC: A three-directions spectral–spatial convolution neural network for hyperspectral image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 377–388, 2021.

[26] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.

[27] C. Zhao, H. Cheng, and S. Feng, "A spectral–spatial change detection method based on simplified 3-D convolutional autoencoder for multitemporal hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] X. Zhang et al., "SMD-Net: Siamese multi-scale difference-enhancement network for change detection in remote sensing," *Remote Sens.*, vol. 14, no. 7, p. 1580, Mar. 2022.

[29] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[30] Y. Yang, J. Qu, S. Xiao, W. Dong, Y. Li, and Q. Du, "A deep multiscale pyramid network enhanced with spatial–spectral residual attention for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5525513.

[31] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518113.

[32] X. Zhang, S. Chen, P. Zhu, X. Tang, J. Feng, and L. Jiao, "Spatial pooling graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521315.

[33] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519912.

[34] X. Li, Z. Yuan, and Q. Wang, "Unsupervised deep noise modeling for hyperspectral image change detection," *Remote Sens.*, vol. 11, no. 3, p. 258, Jan. 2019.

[35] Q. Li et al., "Unsupervised hyperspectral image change detection via deep learning self-generated credible labels," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9012–9024, 2021.

[36] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[37] C. R. Chen, Q. Fan, and R. Panda, "CrossViT: Cross-attention multi-scale vision transformer for image classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 347–356.

[38] X. He, Y. Chen, and Z. Lin, "Spatial-spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[39] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[40] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lucic, and C. Schmid, "ViViT: A video vision transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6816–6826.

[41] Z. Liu et al., "Video Swin transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 3192–3201.

[42] R. Girdhar, J. J. Carreira, C. Doersch, and A. Zisserman, "Video action transformer network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 244–253.

[43] D. Neimark, O. Bar, M. Zohar, and D. Asselmann, "Video transformer network," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 3156–3165.

[44] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[45] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[46] J. Yu, J. Li, Z. Yu, and Q. Huang, "Multimodal transformer with multi-view visual representation for image captioning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 12, pp. 4467–4480, Dec. 2020.

[47] M. Cornia, M. Stefanini, L. Baraldi, and R. Cucchiara, "Meshed-memory transformer for image captioning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10575–10584.

[48] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2844–2853.

[49] I. Misra, R. Girdhar, and A. Joulin, "An end-to-end transformer model for 3D object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 2886–2897.

[50] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[51] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," 2022, *arXiv:2201.01293*.

[52] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

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[53] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.
[54] W. Dong et al., "Abundance matrix correlation analysis network based on hierarchical multihead self-cross-hybrid attention for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501513.
[55] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[56] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," 2016, *arXiv:1606.08415*.
[57] G. H. Rosenfield and K. Fitzpatrick-Lins, "A coefficient of agreement as a measure of thematic classification accuracy," *Photogram. Eng. Remote Sens.*, vol. 52, no. 2, pp. 223–227, 1986.
[58] M. D. Zeiler, "ADADELTA: An adaptive learning rate method," 2012, *arXiv:1212.5701*.
[59] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[60] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3141–3149.
[61] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.



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